

Qns	Answer	Marks
4(a)	By Principle of Conservation of Linear Momentum $(0.50)(2.0) + (0.20)(0) = (0.50)(v_1) + (0.20)(v_2)$ [1]	C1
	By Relative speed of approach = relative speed of separation $u_1 - u_2 = v_2 - v_1$ $2.0 - 0 = v_2 - v_1$ [2]	C1
	Solve equations [1] and [2], $v_2 = 2.857 = 2.9 \text{ m s}^{-1}$	A1
4(b)	By Principle of Conservation of Energy, Loss in kinetic energy = Gain in Gravitational Potential Energy $\frac{1}{2}mv^2 - 0 = mgh$ $\frac{1}{2}(2.857^2) = 9.81h$ $h = 0.416 \text{ m}$	C1
	Consider maximum angle $h = L - L \cos \theta = L(1 - \cos \theta)$ $0.416 = 1.5(1 - \cos \theta)$ $\theta = 43.7^\circ$	C1 A1
4(c)	There is an external resultant force acting on the bob provided by the tension in the string and weight .	B1

